

Detailed Summary

“E-commerce Market Analysis and Product Clustering for Retail Inventory Optimization” by Haley Banks, Jessie Ye, Kevelly Batista Diaz, and Saeed Ehsanfar (Fall 2025) outlines an exploratory data analysis and data mining project conducted on the onlineRetail2 transaction database. The primary objective of the investigation was to model customer purchasing behaviors and extract patterns to guide data-driven decisions for future retail inventory control and targeted sales strategies.

Dataset Diagnostics & Preprocessing

The primary raw transactional matrix was acquired from the UC Irvine Machine Learning Repository and tracks retail line items spanning from December 2010 to December 2011.

Data Cleansing Partition: While the complete external source database consists of 541,910 rows, the group partition focused on a ~400K row dataset. Data auditing revealed 135,080 entries with missing values (NA) and 8,905 transactions with negative volume quantities, which were dropped from the pipeline. The final cleaned analysis database settled at 397,925 structural rows.

Descriptive Statistics: Initial profiling established that the three highest revenue-generating countries were the United Kingdom, the Netherlands, and Eire (Ireland), respectively. Baseline system averages revealed an average of 13.02 units sold per item line, an average item price of 3.12, and an average volume of 26.45 items per unique invoice basket.

Mining Association Rules (Market-Basket Analysis)

The team executed a market-basket affinity analysis to identify co-occurrence rules between distinct items using the Apriori algorithm.

The Baseline Model: Line items were consolidated into 18,536 unique transactional invoice baskets. Running the Apriori pass with a minimum threshold filter of 1% support yielded 275 active association rules. A notable two-item rule emerged showing a powerful bond between pink and green plates; purchases containing a pink plate also contained a green plate in 90.2% of instances (confidence), showing a high conviction score of 10.034.

Purchase Intensity (Binned Quantities): To determine if purchase volumes influenced product associations, transaction quantities were segregated into five equal intensity bins. Re-running the Apriori algorithm across these intensity attributes generated 18 refined rules, proving that specific transaction scales correlate highly (e.g., small orders of pink bunting co-occur 47.48% more frequently with small orders of blue bunting than by chance).

Economic Complementarity: An aggregation of daily quantities and average prices for the plate pairs revealed a notable consumer behavioral trend: as the price of the green plate scaled up, the consumption quantity of pink plates increased, indicating that despite their initial co-occurrence affinity, they can act as market substitutes under fluctuating pricing parameters.

Text Clustering Analysis

To identify latent categories among the inventory space, the team deployed an unsupervised K-means clustering algorithm from scikit-learn (`sklearn.cluster`) using standard Euclidean distance calculations.

Feature Engineering Matrix: Text summaries from the Description column were transformed into numerical arrays using Term Frequency-Inverse Document Frequency (TF-IDF) feature vectors, establishing a processed matrix shape of 397,925 rows and 2,031 vocabulary term columns.

Cluster Assignments: The inventory space was partitioned into 5 thematic clusters:

- Cluster 0 (53,820 items): Kids' crafts, baking sets, and pantry cases.
- Cluster 1 (13,810 items): Hanging candle holders and T-light glassware.
- Cluster 2 (266,512 items): A massive general goods category dominated by terms like "bag", "red", and "retrospot".
- Cluster 3 (36,507 items): Retro, vintage, and Christmas-themed items, heavily anchored by high TF-IDF scores for the words "vintage" and "christmas".
- Cluster 4 (27,276 items): Pink-themed assets, clocks, and alarm items.

We concluded our project with definitive business recommendations, advising retail managers to bundle the highly associated pink and green plates with monetary discounts, coordinate matching inventory ordering levels during restocks to prevent supply disruptions, and account for seasonality patterns indicated by peak purchasing periods to preserve storage spaces and optimize corporate cash flow.